Software Requirements Specification

For

ShareHub – Community Marketplace

Version 1.0

Prepared by Dristy Chowdhury (ID: 4325105101140)

Raoful Hashem(ID: 4325105101174)

Nusrat Jahan Maria(ID:4325105101162)

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to describe the requirements, functionality, scope and overall behavior of ShareHub – Community Marketplace. ShareHub is a web-based platform where users can Sell, Rent, Share (free) or Borrow (lend) items within a community. This document gives the project team, developers and testers a common understanding of what the system must do and how users will interact with it.

1.2 Document Conventions

The terms “shall” and “must” indicate mandatory requirements. The term “should” indicates a recommended requirement or behavior.

1.3 Intended Audience and Reading Suggestions

This document is intended for project team members, developers, testers and teachers who evaluate the project. Developers and testers should read Sections 3 and 4 in detail. Other readers can read Sections 1 and 2 for an overview of the project, then Section 4.2 for the main features.

1.4 Product Scope

ShareHub is a web-based system that connects item owners with people who need those items. Owners list an item for Sale, Rent, Share (free) or Borrow (lend). Other users search, filter and sort the listings, save items to a wishlist or cart, and contact the owner through WhatsApp to complete the deal. Unlike a traditional marketplace, ShareHub combines buying, renting, sharing and borrowing in one platform.

The following features are outside the scope of the current version and are planned as future enhancements:

1. Request management with First-Come, First-Served (FCFS) waiting queue
2. Due-date, return and overdue management
3. Online payment
4. Email or SMS notifications
5. User verification and trust badges
6. In-app messaging and transaction history
7. Editing and deleting a listing

1.5 References

1. Software Engineering – Ian Sommerville, Pearson.
2. Software Engineering: A Practitioner’s Approach – Roger S. Pressman, McGraw-Hill Education.
3. IEEE/ISO/IEC SRS Guidelines.

4. MDN Web Docs: HTML, CSS and JavaScript – Mozilla Developer Network.
5. PHP Manual and MySQL Reference Manual – php.net, mysql.com.
6. Software Requirements: Practical Techniques for Requirements Engineering – Karl E. Wiegers, Joy Beatty.

2. Overall Description

2.1 Product Perspective

ShareHub is a standalone web application with three parts: the browser interface (HTML, CSS, JavaScript), a PHP server layer running on Apache, and a MySQL database. The browser sends requests to PHP files, which read or write the database and return the result in JSON format. The only external service is WhatsApp, which is opened through a chat link when a user contacts a seller.

2.2 Product Functions

1. Register: for visitors
2. Login: for users
3. Logout: for users
4. Add Product Listing (Sell, Rent, Share, Borrow): for sellers and lenders
5. Upload Product Image: for sellers and lenders
6. View Products: for all users
7. Search Products: for all users
8. Filter Products by Category: for all users
9. Sort Products by Price: for all users
10. View Product Details: for all users
11. Add or Remove Wishlist Item: for signed-in users
12. View Wishlist: for signed-in users
13. Add to Cart: for signed-in users
14. Change Cart Quantity and Remove Item: for signed-in users
15. View Cart Total: for signed-in users
16. Contact Seller through WhatsApp: for signed-in users

17. Checkout (information only): for signed-in users

2.3 User Classes and Characteristics

- **Visitor** – A user who has not signed in. A visitor can view, search, filter and sort products, view product details, and register or log in.
- **Buyer / Borrower (Resource Seeker)** – A registered user who finds items. Functions: login, search, wishlist, cart, contact seller, logout.
- **Seller / Lender (Resource Owner)** – A registered user who lists items. Functions: login, add product listing with image, price or borrow period, contact number, logout.

One registered account can act as both buyer and seller.

2.4 Operating Environment

ShareHub runs as a web application on computers, laptops, tablets and smartphones using modern browsers such as Google Chrome and Microsoft Edge. An Internet connection is required. The server runs on Apache with PHP 8 and a MySQL (MariaDB) database, for example through XAMPP. The system is developed using HTML, CSS, JavaScript and PHP.

2.5 Design and Implementation Constraints

a. The system provides four listing types: Sell, Rent, Share and Borrow.

b. Uploaded images must be JPG, PNG, WEBP or GIF and must not exceed 5 MB.

c. Seller contact is done through WhatsApp; online payment is not supported in this version.

d. The system needs an Internet connection and a modern browser.

e. As a student project, large-scale traffic management and advanced features are not included.

2.6 User Documentation

Simple digital guidance will explain registration, login, listing a product, searching and filtering products, using the wishlist and cart, and contacting a seller. The guidance will be easy to follow for both sellers and buyers.

2.7 Assumptions and Dependencies

- Users provide accurate account and product information, including a valid phone number for contact.
- Apache and MySQL are running on the server.

- Seller contact depends on the availability of WhatsApp.
- Demo products may use images hosted on external websites; if those are unavailable, a placeholder image is shown.

3. External Interface Requirements

3.1 User Interfaces

1. **Home Page:** Navigation bar, search bar, wishlist and cart buttons, sign-in and register buttons, category buttons, and the product list.
2. **Sign In:** Allows registered users to log in with email and password. An error message is shown for incorrect information.
3. **Register:** Allows new users to create an account with name, email, password and password confirmation.
4. **List Product:** Allows a signed-in user to add a product with name, category, listing type, price or borrow period, image, location, phone number and description.
5. **Product Details:** Shows image, name, category, price, listing type, location, owner name and description, with Wishlist, Cart and Contact Seller buttons.
6. **Search, Filter and Sort:** Search bar, category buttons and a price sort menu.
7. **Wishlist:** Shows the products saved by the user.
8. **Cart:** Side panel showing items, quantity controls and the total price.

3.2 Software Interfaces

The browser interface is built with HTML, CSS and JavaScript. The server layer uses PHP and connects to the MySQL database through mysqli. The database contains four tables: users, products, wishlist and cart_items. JavaScript communicates with PHP using fetch requests and JSON. WhatsApp is used through a chat link to contact the seller.

3.3 Communications Interfaces

1. The system requires an Internet connection for communication between the user and the web application.
2. Data is transferred between the browser and the server using HTTP (HTTPS when deployed online).
3. Product images are uploaded using multipart form data; other requests use JSON.
4. A session cookie keeps the user signed in.

5. The system updates product, wishlist and cart information after each action so that the displayed data stays accurate.

4. System Features

4.1 User Registration and Login

This is a High Priority feature, because users need an account to list products, use the wishlist and cart, and contact sellers.

1. The visitor selects Register and enters name, email and password.
2. The system validates the information, saves the account with a hashed password and signs the user in.
3. A returning user enters email and password on the Sign In page.
4. The system verifies the information and shows the user's name in the header.
5. If the information is invalid, the system displays an error message.

4.2 Functional Requirements

REQ-1: Register

- **User:** Visitor
- **Input:** Name, email address, password (minimum 6 characters) and password confirmation
- **Output:** Account is created, the user is signed in and the name is shown in the header
- **Alternative flow(s):** If the email is invalid or already registered, or the passwords do not match, the system shows an error message

REQ-2: Login

- **User:** Registered user
- **Input:** Registered email address and password
- **Output:** User is signed in, the name is shown, and the saved wishlist and cart are loaded
- **Alternative flow(s):** If the email or password is incorrect, the system shows an error message

REQ-3: Logout

- **User:** Signed-in user

- **Input:** Click the name button and confirm
- **Output:** The session ends and the Sign In and Register buttons are shown
- **Alternative flow(s):** None

REQ-4: Add Product Listing

- **User:** Signed-in user (seller or lender)
- **Input:** Product name, category, listing type (Sell, Rent, Share or Borrow), price (for Sell and Rent), rent period (for Rent), maximum borrow days (for Borrow), image, location, phone number and description
- **Output:** The product and its image are saved and the listing appears on the home page
- **Alternative flow(s):** If a field is missing, the price is zero for Sell or Rent, or the image is larger than 5 MB or not a valid image, the system shows an error message. If the user is not signed in, the sign-in window opens

REQ-5: View Products

- **User:** All users
- **Input:** Open the home page
- **Output:** Product cards showing image, listing type, category, name, description, price (FREE for Share and Borrow), rating and location
- **Alternative flow(s):** If there are no products, the system shows the message No products found

REQ-6: Search Product

- **User:** All users
- **Input:** Keyword typed in the search bar
- **Output:** Products whose name, category, location, listing type or description match the keyword
- **Alternative flow(s):** If no product matches, the system shows a suitable message

REQ-7: Filter by Category

- **User:** All users
- **Input:** Click a category button

- **Output:** Only products of that category are shown and the selected category is highlighted
- **Alternative flow(s):** If the category has no products, the system shows a suitable message

REQ-8: Sort Products by Price

- **User:** All users
- **Input:** Select Price: Low to High or Price: High to Low
- **Output:** Products are shown in the selected order
- **Alternative flow(s):** None

REQ-9: View Product Details

- **User:** All users
- **Input:** Click a product card
- **Output:** Image, name, category, price, listing type, location, owner name, description and rating
- **Alternative flow(s):** None

REQ-10: Manage Wishlist

- **User:** Signed-in user
- **Input:** Click the heart button on a product, or click Wishlist in the header
- **Output:** The product is added to or removed from the saved wishlist, the wishlist count is updated and the My Wishlist view shows the saved products
- **Alternative flow(s):** If the user is not signed in, the sign-in window opens. If the wishlist is empty, the system shows the message Your wishlist is empty

REQ-11: Manage Cart

- **User:** Signed-in user
- **Input:** Click Add to Cart, use the plus and minus buttons, or click the remove button
- **Output:** The quantity is saved, and the cart count and total price are updated. For rent items, the quantity means the number of rent periods

- **Alternative flow(s):** If the user is not signed in, the sign-in window opens. If the quantity becomes zero, the item is removed from the cart

REQ-12: Contact Seller

- **User:** Signed-in user
- **Input:** Click Contact Seller on the product details window
- **Output:** A WhatsApp chat opens with the seller's number and a message containing the product name
- **Alternative flow(s):** If the user is not signed in, the sign-in window opens. If the listing has no phone number, the system shows a message

REQ-13: Checkout

- **User:** Signed-in user
- **Input:** Click Continue to Checkout in the cart
- **Output:** The system tells the user to open each product and contact the seller to complete the deal, because online payment is not available in this version
- **Alternative flow(s):** If the cart is empty, the system shows the message Your cart is empty

4.3 Non-Functional Requirements

1. **Performance:** The system should respond to search, viewing and cart operations within a reasonable time under normal usage.
2. **Usability:** The interface should be simple and clear so that users with basic web knowledge can use it easily.
3. **Reliability:** The system should perform its main functions consistently and keep product, wishlist and cart data accurate.
4. **Security:** Passwords must be stored as hashes, database queries must use prepared statements, user text must be escaped before display, uploaded files must be checked as real images, and login must use a server session.
5. **Availability:** The system should be available whenever the server is running and the user has a stable Internet connection.
6. **Maintainability:** The system should have an organized structure (separate HTML, CSS, JavaScript and PHP files) so that it is easy to modify and extend.

7. **Compatibility:** The system should work on common operating systems and modern browsers such as Google Chrome and Microsoft Edge.
8. **Scalability:** The system should allow future addition of more users, products, categories and features such as payment and notifications.
9. **Data Integrity:** Each email must be unique, and related data must stay consistent using database keys, so duplicate or invalid records are prevented.
10. **Responsiveness:** The interface should work properly on computers, laptops, tablets and smartphones.
11. **Privacy:** Seller phone numbers should be visible only to signed-in users, and passwords must never be shown or stored in plain text.
12. **Error Handling:** The system should display clear messages when information is invalid or an operation cannot be completed.